

JEE MAIN REVISION MASTER: IONIC EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 140 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Ionic Equilibrium - Core Application	140 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Ionic Equilibrium - Core Application (Very High Yield — 140 PYQs)

Core Principle: Key problem pattern in Ionic Equilibrium.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given is a concentrated solution of a weak electrolyte $A_{(x)}B_{(y)}$ of concentration 'c' and dissociation constant 'K'. The degree of ..." [Ans: N/A]
- [2026 Evening Shift] Q#1 (MCQ): "20 mL of a solution of acetic acid required 28.4 mL of 0.1 M NaOH for its neutralization. A solution (X) was prepared by mixing 20 mL of the..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Consider a weak base 'B' of $pK_{(b)}=5.699$. 'x' mL of 0.02 M HCl and 'y' mL of 0.02 M weak base 'B' are mixed to make 100 mL of a..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

The first and second ionization constants of H_2X are 2.5×10^{-8} and 1.0×10^{-13} respectively. The concentration of X^{2-} in 0.1M H_2X solution is _____ $\times 10^{-15}M$. (Nearest Integer)

Correct Answer: 100

Solution Analysis: For the diprotic acid H_2X (total concentration $C=0.1M$): $K_{(a1)}=2.5 \times 10^{-8}, K_{(a2)}=1.0 \times 10^{-13}$ 1) Find $[H^+]$ (second dissociation is negligible for $[H^+]$): $[H^+] \approx \sqrt{(K_{(a1)}C)} = \sqrt{(2.5 \times 10^{-8})(0.1)} = \sqrt{2.5 \times 10^{-9}} = 5.0 \times 10^{-5}M$ 2) Use diprotic distribution for $[X^{2-}]$: $[X^{2-}] = C \cdot \frac{K_{(a1)}K_{(a2)}}{[H^+]^2 + K_{(a1)}[H^+] + K_{(a1)}K_{(a2)}}$ Compute key terms: $\frac{K_{(a1)}K_{(a2)}}{[H^+]^2} = \frac{(2.5 \times 10^{-8})(1.0 \times 10^{-13})}{(5 \times 10^{-5})^2} = \frac{2.5 \times 10^{-21}}{2.5 \times 10^{-9}} = 10^{-12}$